

Paper 3A: Further Pure Mathematics 1

Topics	What students need to learn:		
	Content		Guidance
1 Further Trigonometry	1.1	The t-formulae	The derivation and use of $\sin \theta \equiv \frac{2t}{1+t^2}, \cos \theta \equiv \frac{1-t^2}{1+t^2},$ $\tan \theta \equiv \frac{2t}{1-t^2}, \text{ where } t = \tan \frac{\theta}{2}$
	1.2	Applications of t-formulae to trigonometric identities	E.g. show that $\frac{1 + \operatorname{cosec} \theta}{\cot \theta} \equiv \frac{1 + \tan \frac{\theta}{2}}{1 - \tan \frac{\theta}{2}}$
	1.3	Applications of t-formulae to solve trigonometric equations	E.g. the solution of equations of the form $a \cos x + b \sin x = c$